TEACHER MARK SCHEME

Recurring decimals and exact fractions

Companion to the two-hour student tutorial. Original questions. Not an IB mark scheme and not a reproduction of a specimen or past paper.

Where this sits in the eAssessment

The mathematics skills framework, standard *and* extended, lists recurring decimals among the forms of numbers, asks for transformation between forms, and asks students to classify numbers, including recurring decimals. The published notation list uses dots over the first and last repeating digits, not a bar. The on-screen examination assesses Criteria A–D. This lesson trains the moves those items reward:

- **A** — choose the multiplier from the block length, carry out the subtraction, simplify.
- **B** — cases, a general rule in terms of k, a fresh verification, and a justification. A table alone does not meet the top of the criterion.
- **C** — dot notation, a defined variable, and a concluding sentence.
- **D** — an exact concentration, the direction of a rounding error, and a reason that matters in the calculation that follows.

Specimen and past papers were used only to confirm those demands. No official item has been copied. The repeat-length investigation that appears in some other international papers is a different task; B1–B5 is an original nines-family investigation.

Every question on pages 2–11 of the student booklet is core. Extended candidates need it as well. E1–E4 are optional and are not required by the standard route. There is no Standard-only replacement page, because this topic is already in the standard framework.

Marks in the tutorial: diagnostic 6, practice A 11, mixed M 11, both directions P 8, investigation B 14, context T 8, checkpoint C 10, exit X 5. Enrichment is marked separately and is outside the 120 minutes.

Accept any equivalent simplified fraction. **Do not accept** a correct decimal in place of a requested fraction, or a recurring bar if the student was told to use dots, unless you have explicitly allowed both notations in class. If both notations are in use, the value must still be right.

Diagnostic | 6 marks

D1 $\dfrac{375}{1000} = \dfrac{3}{8}$. [1]
Award the mark for 3/8 only. 375/1000 without simplification: 0.

D2 The block 27 repeats, in that order: $0.272727\ldots$ [1]
"2 and 7" with no order is not enough. "Only 7" is the mixed-decimal misreading: 0.

D3 Rational, because the digit 4 recurs, so the number can be written as a fraction (4/9). [1]
"Rational" with no reason: 0 if you are using this as a classification item; accept the reason "it is a recurring decimal".

D4 $\dfrac{16}{100} = \dfrac{4}{25}$. [1]

D5 Only 6. The digit 1 does not repeat. [1]
This is the item that predicts success on M4 and C2.

D6 $0.1\dot{8}$. [1]
$0.1\dot{8}$ is $0.1888\ldots$ and scores 0. $0.\overline{18}$ may be accepted if the class has not yet met dot notation; otherwise ask for the dots.

Routing. Missed D1 or D4: two minutes on place value before example 1. Missed D2, D5 or D6: say three decimals aloud before example 3. D3 wrong: return to the definition on page 3 before the investigation.

Practice A | 11 marks

A1 $x = 0.\dot{4}$, $10x = 4.\dot{4}$, $9x = 4$, $x = \dfrac{4}{9}$. [1]

A2 $x = 0.\dot{1}\dot{8}$, $100x = 18.\dot{1}\dot{8}$, $99x = 18$, $x = \dfrac{18}{99} = \dfrac{2}{11}$. [2]

M1 for $99x = 18$ or equivalent. A1 for 2/11. 18/99 alone scores the method mark only.

A3 The block is 06, two digits, even though it starts with zero.

$$x = 0.\dot{0}\dot{6}, \quad 100x = 6.\dot{0}\dot{6}, \quad 99x = 6, \quad x = \frac{6}{99} = \frac{2}{33}.$$

[2]

M1 for recognising a two-digit block (multiplier 100, or denominator 99). A1 for 2/33. Treating it as $0.0666\ldots$ is a notation error and scores 0.

A4 $0.\dot{4} = \dfrac{4}{9}$, so $2.\dot{4} = 2 + \dfrac{4}{9} = \dfrac{22}{9}$. [2]

M1 for splitting the integer, or for $10x - x = 22$ with $x = 2.\dot{4}$. A1 for 22/9. $2\dfrac{4}{9}$ is acceptable.

A5 $x = 0.\dot{2}\dot{7}$, $100x = 27.\dot{2}\dot{7}$, $99x = 27$, $x = \dfrac{27}{99} = \dfrac{3}{11}$. [2]

The question says "show that", so the simplification must be visible. A correct answer with no working does not score both marks.

A6 One decimal *place* would mean the digit stops, giving tenths. Here 3 repeats, so there are infinitely many decimal places and the correct comparison is $10x - x$. $\dfrac{3}{10} = 0.3$, which is not $0.\dot{3}$. The correct fraction is $\dfrac{1}{3}$. [2]

1 mark for the contrast between terminating and recurring. 1 mark for 1/3.

Mixed recurring | 11 marks

M1 $x = 0.2\dot{7} = 0.2777\ldots$

$$\begin{aligned} 10x &= 2.777\ldots \\ 100x &= 27.777\ldots \\ 90x &= 25 \\ x &= \frac{25}{90} = \frac{5}{18}. \end{aligned}$$

[3]

M1 for a correct pair of multipliers. M1 for $90x = 25$ or equivalent. A1 for 5/18. 25/90 scores the two method marks only.

M2 $x = 0.1\dot{6}$

$$\begin{aligned} 10x &= 1.666\ldots \\ 100x &= 16.666\ldots \\ 90x &= 15 \\ x &= \frac{15}{90} = \frac{1}{6}. \end{aligned}$$

[2]

Alternative: $10x - x = 1.5$, so $9x = 3/2$ and $x = 1/6$. That is valid. Losing the fraction in 1.5/9 and writing $0.166\ldots$ as the "answer" does not complete a "show that".

M3 $x = 0.12\dot{3} = 0.12333\ldots$

$$\begin{aligned} 100x &= 12.333\ldots \\ 1000x &= 123.333\ldots \\ 900x &= 111 \\ x &= \frac{111}{900} = \frac{37}{300}. \end{aligned}$$

[3]

The non-repeating part has *two* digits, so the first multiplier is 100, not 10. Using 10 and 100 is the error to look for; it solves $0.1\dot{2}\dot{3}$ or a different misreading and does not score the accuracy mark.

M4 $0.1\dot{6} = \frac{16}{99}$. [1]

$0.1\dot{6} = \frac{1}{6}$. [1]

$\frac{1}{6} = \frac{15}{90}$ and $\frac{16}{99} = \frac{16}{99}$. Cross-multiply: $15 \times 99 = 1485$ and $16 \times 90 = 1440$, so $\frac{1}{6} > \frac{16}{99}$. Thus $0.1\dot{6}$ is larger. [1]

A decimal comparison is acceptable if it identifies $0.1666\ldots$ and $0.1616\ldots$ without rounding them to the same value.

Both directions | 8 marks

P1 $0.8\dot{3}$. [1]

$0.\dot{8}\dot{3}$ is $0.838383\ldots$ and scores 0. 0.833 without a dot scores 0.

P2 $0.\dot{1}\dot{8}$. [1]

P3

a) $\frac{9}{40}$: $40 = 2^3 \times 5$, terminating. [1]

b) $\frac{7}{15}$: $15 = 3 \times 5$, the factor 3 remains, recurring. [1]

c) $\frac{5}{12}$: $12 = 2^2 \times 3$, recurring. [1]

d) $\frac{3}{8}$: $8 = 2^3$, terminating. [1]

The fraction must be considered in lowest terms. These four already are. Award the mark for the correct word *with* a prime-factor reason. The word alone, in a lesson whose objective is the test, scores 0.

P4 $\frac{9}{40} = \frac{225}{1000} = 0.225$. [1]

It terminates, which agrees with P3a because the denominator has no prime factor other than 2 or 5. [1]

Investigation B | 14 marks

B1 $0.\dot{1}$, $0.\dot{2}$, $0.\dot{4}$, $0.\dot{8}$. [2]
All four correct: 2. Three correct: 1. Two or fewer: 0.

B2 $\dfrac{12}{99} = 0.\dot{1}\dot{2}$ and $\dfrac{7}{99} = 0.\dot{0}\dot{7}$. [2]
The leading zero in 07 must appear in the description or the notation. $0.\dot{7}$ for the second value is incorrect.

B3 For a one-digit block the denominator is 9; for a two-digit block it is 99. In general the denominator is a string of nines, one nine for each digit in the block. The numerator is the block itself. What stays the same is the use of nines only; what changes is how many nines, matching k. [2]
1 mark for linking the number of nines to the length of the block. 1 mark for identifying the numerator as the repeating block. A list of examples with no general statement scores 0.

B4 $0.\overline{N} = \dfrac{N}{10^k - 1}$, or N over a number made of k nines. [3]
1 mark for k nines or $10^k - 1$. 1 mark for the numerator N, including the instruction that a leading zero still counts in k. 1 mark for a sentence that states the rule for every purely recurring decimal, not only the examples. "For every positive integer k" is the sentence to listen for.

B5 (a) $k = 3$, $N = 142$, so $\dfrac{142}{999}$. $\gcd(142, 999) = 1$, because $142 = 2 \times 71$ and $999 = 3^3 \times 37$. The fraction is already in simplest form. [2]
1 mark for 142/999. 1 mark for a correct statement that it does not simplify, with a reason better than "they look coprime".

(b) Let $x = 0.\dot{1}4\dot{2}$. Then $1000x = 142.\dot{1}4\dot{2}$ and

$$1000x - x = 142, \qquad 999x = 142, \qquad x = \frac{142}{999}.$$

Multiplying by 10^3 shifts exactly one three-digit block, so the subtraction removes the tail and leaves the block as an integer. The denominator is $10^3 - 1 = 999$, which is the rule. [3]
M1 for the multiplier 1000. M1 for $999x = 142$. A1 for the concluding link to $10^k - 1$.

Criterion B note. A student who completes B1–B2 only is still at the level of examples. B4 without B5 can be a rule that has been noticed but not justified. The justification mark in B5(b) is the one that corresponds to explaining *why* the pattern continues, not merely checking one more value.

Context T | 8 marks

T1 $0.1\dot{6} = \dfrac{1}{6}$ mg/L, by the working in M2. [2]

M1 for $90x = 15$ or equivalent. A1 for 1/6, with the unit if the student is writing a sentence. The unit is expected in a context answer; do not withhold the accuracy mark solely for omitting "mg/L" if the fraction is clearly the concentration.

T2

$$0.17 - \frac{1}{6} = \frac{17}{100} - \frac{1}{6} = \frac{51}{300} - \frac{50}{300} = \frac{1}{300}.$$

The recorded value is too high, by $\dfrac{1}{300}$ mg/L. [3]

M1 for writing 0.17 as 17/100. M1 for a correct subtraction with a common denominator. A1 for 1/300 and "too high".

T3 Mass $= \dfrac{1}{6} \times 90 = 15$ mg. [1]

Using 0.17 gives $0.17 \times 90 = 15.3$ mg, which is 0.3 mg too high. [1]

The error is multiplied by 90, so a small excess in the concentration becomes a larger excess in the mass. The exact recurring form avoids that. [1]

Accept $90 \times \dfrac{1}{300} = \dfrac{90}{300} = \dfrac{3}{10}$ mg as the excess mass. The explanation mark requires the multiplication of the error, not only "rounding is inaccurate".

Checkpoint C | 10 marks

C1 $x = 0.\dot{4}\dot{5}$, $100x = 45.\dot{4}\dot{5}$, $99x = 45$, $x = \dfrac{45}{99} = \dfrac{5}{11}$. [2]

M1 for $99x = 45$. A1 for $5/11$.

C2 $x = 0.4\dot{5} = 0.4555\ldots$

$$\begin{aligned} 10x &= 4.555\ldots \\ 100x &= 45.555\ldots \\ 90x &= 41 \\ x &= \frac{41}{90}. \end{aligned}$$

[3]

This does not simplify: $\gcd(41, 90) = 1$. M1 for multipliers 10 and 100. M1 for $90x = 41$. A1 for $41/90$. The discrimination with C1 is the point. A student who gives $5/11$ for both has not read the second dot pattern.

C3 $0.\dot{2}\dot{7}$ is rational because its decimal recurs, so the shift method produces the fraction $3/11$. A decimal that neither terminates nor recurs cannot be written as a ratio of integers, so it is irrational. [2]

1 mark for a correct reason that $0.\dot{2}\dot{7}$ is rational. 1 mark for the non-terminating, non-recurring condition. Naming π or $\sqrt{2}$ as an example may support the second mark but is not required.

C4 Let $x = 0.\dot{9} = 0.999\ldots$ Then $10x = 9.999\ldots$ and $10x - x = 9$, so $9x = 9$ and $x = 1$. [3]

M1 for $10x$. M1 for $9x = 9$. A1 for the conclusion $x = 1$, not "approximately 1".

If a student writes "it gets closer but never arrives", that contradicts the algebra and does not score the accuracy mark. E4 returns to this.

Exit X | 5 marks

X1 $\dfrac{6}{9} = \dfrac{2}{3}$. [1]
6/9 alone: 0.

X2 $0.\dot{1}\dot{2} = \dfrac{12}{99} = \dfrac{4}{33}$, so

$$3.\dot{1}\dot{2} = 3 + \frac{4}{33} = \frac{99}{33} + \frac{4}{33} = \frac{103}{33}.$$

[2]
Or $100x - x = 312 - 3 = 309$ is *not* the right subtraction for $x = 3.\dot{1}\dot{2}$, because $100x = 312.\dot{1}\dot{2}$ and $100x - x = 309$, so $99x = 309$ and $x = 309/99 = 103/33$. That full method scores both marks. The slip 312/99 scores 0.

X3 15/90 and 1/6 are equal, but 1/6 is in simplest form. A final answer should have no common factor greater than 1 in the numerator and denominator, unless the question asks for an unsimplified stage. It is also easier to compare and to reuse. [2]
1 mark for naming simplest form or dividing by 15. 1 mark for a reason about communication or later use.

Enrichment

E1 Let x be purely recurring, with block N of k digits.

$$10^k x = N + x,$$

because the shift moves exactly one block to the left of the decimal point and the tail is x again. Hence

$$(10^k - 1)x = N, \qquad x = \frac{N}{10^k - 1}.$$

[4]

1 mark for defining the block length. 1 mark for the multiplier 10^k. 1 mark for $10^k x - x = N$. 1 mark for the conclusion, with N an integer that may include a leading zero in its decimal block but not in its value.

E2 $x = 0.1\dot{4}\dot{2} = 0.1424242\ldots$

$$\begin{aligned} 10x &= 1.424242\ldots \\ 1000x &= 142.424242\ldots \\ 990x &= 141 \\ x &= \frac{141}{990} = \frac{47}{330}. \end{aligned}$$

[4]

M1 for $10x$ and $1000x$. M1 for $990x = 141$. A1 for $141/990$. A1 for $47/330$.
Check: $47/330 = 0.1424242\ldots$

E3

$$\frac{5}{8} = 0.625, \quad \frac{7}{11} = 0.\dot{6}\dot{3} = \frac{63}{99}, \quad \frac{2}{3} = 0.\dot{6}, \quad 0.\dot{6}\dot{7} = \frac{61}{90}.$$

Common denominator 3960:

$$\frac{5}{8} = \frac{2475}{3960}, \qquad \frac{7}{11} = \frac{2520}{3960}, \qquad \frac{2}{3} = \frac{2640}{3960}, \qquad \frac{61}{90} = \frac{2684}{3960}.$$

Ascending order:

$$\frac{5}{8}, \quad \frac{7}{11}, \quad \frac{2}{3}, \quad 0.\dot{6}\dot{7}.$$

[5]

1 mark for 61/90. 1 mark each for a correct comparison that places the next value, or 3 marks for a correct common-denominator or exact-decimal argument that orders all four, plus 1 mark for the ascending list. A calculator order with no exact forms scores at most 1, for the list, and only if the list is correct.

E4 From C4, $0.\dot{9} = 1$ exactly. There is no number "just before" 1 on the real line: between any two reals there is another. The sentence treats a second way of writing 1 as if it were a smaller number. [3]

1 mark for using $0.\dot{9} = 1$. 1 mark for rejecting "just before". 1 mark for a density or "same number, two names" statement.

Mistakes worth rehearsing

What the student does	What to say
Uses 10 for every recurring decimal	"Count the digits under the dots, or from the first dot to the last."
Simplifies $0.0\dot{6}\dot{}$ to $0.\dot{6}$	"A zero inside the block is part of the block."
Stops at 27/99	"Simplest form is part of the answer unless the question says 'show that' and the target is already simplified."
Says $0.\dot{9} \approx 1$	"The subtraction is exact. $9x = 9$ has one solution."
Gives a rounded decimal in T3	"The mass was available exactly. Rounding belongs at the end, and only if a question asks for it."
Writes a rule as 'it is always over 99'	"That was true for the two-digit cases only. Name k."

On-screen rehearsal

Before the next timed paper, ask students to enter three answers in the equation editor: $0.\dot{1}\dot{8}$, $\frac{5}{11}$ and $\frac{41}{90}$. The errors to catch are a dot over one digit of a two-digit block, a fraction bar that does not cover the denominator, and a calculator string such as `0.454545`. None of those is the mathematical answer.